

Table of Content

1. Introduction

- 1.1. Research Gap
- 1.2. Need for the Study

2. Literature Review

- 2.1. Theoretical foundation
- 2.2. Uses & Gratification (U&G) Theory
- 2.3. Hyper-Personalisation in Digital Marketing
- 2.4. Constructs of Hyper-Personalisation
- 2.5. Privacy, Trust and Ethical Considerations
- 2.6. Customer Engagement in Hyper-Personalised Environments

3. Development of the Conceptual Framework

4. Research Hypothesis

- 4.1. Relevance and Customer Value Gratification
- 4.2. Usefulness and Customer Value Gratification
- 4.3. Individualised Experience and Customer Value Gratification
- 4.4. Real Time Personalisation and Customer Value Gratification
- 4.5. Emotional Connect and Customer Value Gratification
- 4.6. Convenience and Customer Value Gratification

5. Research Methodology

- 5.1. Research Design
- 5.2. Sampling Design
- 5.3. Measures
- 5.4. Data Collection Procedure

6. Results

- 6.1. Common Method Bias
- 6.2. Measurement Model and Reliability Analysis
- 6.3. Convergent and discriminant validity analysis
- 6.4. Structural model analysis
- 6.5. Structural Model Results
- 6.6. Mediation analysis

7. Discussions

- 7.1. Summary of Findings
- 7.2. Theoretical Implications
- 7.3. Practical Implications
- 7.4. Limitations and Directions for Future Research

8. Conclusions

9. References

1. Introduction

Businesses are rethinking their products around data-driven personalised experiences as a result of digital transformation, which has altered how they interact with customers. In order to match today's consumer demands, businesses now rely on AI, machine learning, and sophisticated analytics to understand individual likes and provide pertinent content across digital platforms (Florido-Bétez, 2024; Chen et al. 2023; Jain, Paul & Shrivastava 2021). In order to create more accurate, comprehensive customer profiles that promote consistent customer experiences across channels, these solutions assist brands in collecting vast volumes of structured, behavioural, and transactional data from consumers (Jain et al. 2021). Hyper-personalization is a crucial component of contemporary customer experience design since consumers anticipate more individualised communication as they use more mobile apps, online shopping sites, and digital services (Hyken 2017; Lee & Ko 2020; Alfouzan et al., 2024).

Beyond demographic segmentation, hyper-personalization is a sophisticated type of digital personalisation that uses behavioural grouping, deep algorithms, and real-time data to provide personalised messages and suggestions (Micu et al., 2022; Jain et al., 2021). This is further improved by generative AI tools, which analyse past and present client interactions to dynamically customise offers, content, and interface experiences (Matter & Gado, 2024; CapGemini, 2017). By matching communications with customer motivations, purchase histories, and behavioural patterns, AI-driven customisation enhances engagement and attitudes, according to a number of studies (Sahni et al., 2018; Brodie et al., 2013). Additionally, hyper-personalization enables more integrated omnichannel journeys, where channel engagement and unified data enhance consumer response and retention (Bloom, 2019; Hart, 2016). Many brands view these technologies as strategic assets that improve customer connections and expedite decision-making, rather than just operational tools.

Hyper-personalization, however, necessitates a great degree of openness and confidence. Customers may view personalised material as intrusive if privacy boundaries are unclear or if data practices are not freely disclosed, according to research on personalised advertising (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). According to studies (Tucker, 2014; Chen et al., 2019), intrusive personalisation may cause unfavourable reactions, decrease the effectiveness of advertisements and detract from the overall customer experience. Maintaining user control and protecting data becomes essential for sustaining positive engagement as platforms depend more and more on behavioural tracking and algorithmic predictions. Industry publications also highlight the need for businesses to continuously reorganise and reorganise customer journeys in order to maintain advantageous and manageable customised interactions (KPMG, 2018).

The Uses and Gratifications (U&G) theory offers a helpful perspective to comprehend why customers respond favourably or unfavourably to highly customised experiences. According to U&G, people actively select media and material to meet demands such knowledge, entertainment, convenience, social utility, and emotional fulfilment (Ho & Syu, 2010; UGT 21st Century Review, 2020). Because customers now have more

power over what they choose to interact with in digital environments, research indicates that customised experiences are more effective when they coincide with these goals (Spang, 2017). Users are expecting more adaptive, relevant, and seamless digital experiences as streaming platforms and customised interfaces customise content based on past behaviours and preferences (Why We Watch—U&G Theory, 2021). Hyper-personalized material increases perceived value and fortifies emotional connection when it reinforces these pleasures; when it doesn't, it runs the danger of resistance or disengagement.

Hyper-personalization has become crucial in influencing consumer perceptions, experiences, and decision-making processes due to the quick growth of digital ecosystems and AI-driven marketing technologies. However, businesses in this field still struggle to strike a balance between ethical transparency and customised relevance. Understanding the elements that promote good satisfaction, trust, and engagement is essential for creating successful customised strategies as firms negotiate these complications.

1.1. Research Gap

In order to engage customers and comprehend their changing expectations, behavioural patterns, and personal preferences, retailers and digital platforms are depending more and more on hyper-personalized marketing. This change reflects a larger trend in marketing, where maintaining customer interest and loyalty requires personalised content, tailored offers, and AI-enabled recommendations (Florido-Bópeztez, 2024; Jain et al., 2018; Rane et al., 2023). According to Sahni et al. (2018) and Brodie et al. (2013), brands must now inspire consumers in digital contexts by providing highly customised experiences through a variety of channels in an effort to improve user satisfaction and impact behavioural outcomes like purchase intention, engagement, and continued usage. Similar to this, existing research indicates that when systems match users' goals, motives, and past behaviours, users actively contribute to creating customised encounters (Lusch, Vargo, & Tanniru, 2010; Wang, Pallister, & Foxall, 2006). Personalisation has grown more dynamic, real-time, and data-intensive due to the development of AI, the spread of mobile applications, and omnichannel digital touchpoints. This has greatly increased its impact on consumer decision-making (Chen et al., 2023; Alfouzan et al., 2024).

Despite this growing emphasis, most previous research has focused on the operational advantages for businesses or the technological sophistication of AI-driven personalisation, paying relatively little attention to how customers psychologically interpret and assess these highly personalised experiences. Although studies on AI-mediated content and personalised advertising draw attention to problems like intrusiveness, privacy concerns, and trust, they rarely link these responses to consumer value satisfaction or the wider psychological effects of hyper-personalization (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). Similarly, while the Uses and Gratifications (U&G) framework has been extensively used in digital platforms, mobile applications, and online communities to comprehend motivations like entertainment, social interaction, and information seeking, its application in the particular context of hyper-personalized marketing is still in its infancy (Spang,

2017; Ho & Syu, 2010). Because hyper-personalized interactions frequently combine functional benefits with emotional and experiential gratifications, creating a complex consumer experience that standard U&G studies do not fully capture, this gap becomes increasingly apparent.

Furthermore, rather than looking at how hyper-personalized experiences boost engagement and add to perceived value, research on customer engagement typically concentrates on behavioural reactions or social media interactions. Although engagement has been acknowledged as a crucial mediator that moulds loyalty and fortifies brand relationships, it is still unclear how hyper-personalized marketing strategies either help or hurt engagement (Hollebeek et al., 2019; Davenport, 2023). In a cross-sectional, multi-channel digital context, there is also a conspicuous dearth of empirical research connecting hyper-personalization to consumer value gratification, especially in light of the psychological and sensory gratifications described in U&G theory.

Furthermore, there is still a lack of research on hyper-personalization in India's quickly changing digital consumer ecosystem, which includes rising mobile penetration, e-commerce usage, and AI-driven brand interactions. There is little knowledge about how Indian customers view, negotiate, and get value from hyper-personalized material because the majority of empirical research comes from Western markets or concentrates on particular businesses (Alfouzan et al., 2024; KPMG, 2018). A major scholarly opportunity is highlighted by the lack of comprehensive studies that integrate hyper-personalization, customer involvement, and value gratification via the lens of U&G theory.

By analysing digitally driven hyper-personalized marketing through a consumer-centric lens and concentrating on how various aspects of customisation affect customer value fulfilment and engagement, the current study seeks to close these gaps. This study provides a more thorough academic viewpoint on the phenomena and advances our understanding of the psychological effects of hyper-personalization by using an integrated approach based on U&G theory.

1.2. Need for the Study

Hyper-personalized marketing is now a key part of the customer experience due to the quick growth of AI-driven digital environments. Understanding how customers truly interpret and get value from these personalised touchpoints is crucial as businesses use machine learning, predictive analytics, and unified data systems to customise content and interactions (Florido-Bópeztez, 2024; Chen et al., 2023; Jain et al., 2021). The significance of this study lies in the need for practitioners to have a better understanding of the elements that influence consumer acceptability, satisfaction, and engagement in increasingly customised digital environments (Hart, 2016; Sahni et al., 2018).

Hyper-personalization promises more convenience and relevance, but how well it fits the wants and motives of digital customers will determine how effective it is. According to other studies, customisation only increases

value when it aligns with user expectations, emotional reactions, and behavioural patterns (Brodie et al., 2013; Hollebeek et al., 2019). As cutting-edge technologies like generative AI dynamically modify information in real time, it is becoming more important to look at the psychological effects of these systems rather than just their technical complexity (Matter & Gado, 2024; CapGemini, 2017).

The issues of privacy and intrusiveness, which are often linked to data-driven personalisation, are another driving force behind this study. According to studies (Zhu & Kanjanamekanant, 2020; Ferrara, 2023; Tucker, 2014), when customers perceive a lack of control or understanding about how their data is being used, personalised communication may cause discomfort or unfavourable reactions. Developing tactics that actually improve value satisfaction requires an understanding of where customers draw the line between helpful and intrusive personalisation.

The study makes a theoretical contribution to the field of hyper-personalized marketing by utilising Uses and Gratifications (U&G) theory. While U&G has been used extensively to explain why people use digital media, its use in real-time, AI-driven personalisation is still rather limited (Spang, 2017; Ho & Syu, 2010; UGT 21st Century Review, 2020). Examining these links closes a significant conceptual gap because hyper-personalization directly affects the satisfactions that consumers seek, such as relevance, utility, convenience, and emotional connection.

Lastly, there is a contextual need to examine these dynamics inside the Indian digital environment, where consumer expectations are being reshaped by the quick rise in smartphone usage, e-commerce uptake, and AI-enabled marketing. There is little knowledge about how Indian digital customers assess hyper-personalized experiences because the majority of current research is Western-focused (KPMG, 2018; Alfouzan et al., 2024). Therefore, the purpose of this study is to provide light on how hyper-personalization affects customer value satisfaction and engagement in a market that is diverse and changing quickly.

2. Literature Review

Theoretical viewpoints on digital media consumption, personalised service delivery, data-driven technology, and new research on hyper-personalization are all included in the literature that supports this study. This section explains the scholarly underpinnings of each construct and lays out the foundations that form the conceptual model.

2.1. Theoretical Foundation

It has long been acknowledged that personalisation, in which businesses modify products to better meet consumer wants and increase perceived value, is a catalyst for service innovation. Previous studies on digital transformation reveal that businesses are increasingly using technology, behavioural insights, and unified data systems to rethink customer experiences (Jain et al., 2021; Galindo-Martín et al., 2019). Structured and unstructured data can be captured thanks to digitalisation, facilitating more sophisticated and responsive consumer interactions (Jain et al., 2021; Erevelles, Fukawa & Swayne, 2016). These advancements are in line with more general technological changes in marketing, where AI and machine learning examine behavioural patterns to enable large-scale customisation (Florido-Benítez, 2024; Kumar et al., 2020).

Businesses are depending more and more on consumer data to create customised journeys that offer suggestions, carefully chosen content, and cross-platform adaptability (Bloom, 2019; Hart, 2016). Additionally, research indicates that by providing information that is more pertinent to customers' immediate requirements, customisation enhances decision-making and lessens information overload (Brodie et al., 2013; Hyken, 2017). Hyper-personalization, which incorporates emotional, behavioural, and contextual inputs in real time, is the next phase of service innovation as organisations shift towards more accurate types of digital targeting.

2.2. Uses and Gratifications (U&G) Theory

U&G theory looks at why people use media and what they want from digital encounters. Modern reinterpretations of U&G highlight demands such as information utility, social engagement, convenience, and emotional resonance, which govern individuals' choices across digital channels (Ho & Syu, 2010; U&G in the 21st Century, 2020). As customised digital ecosystems become more prevalent, user satisfaction is more closely linked to how well the system adjusts to user preferences.

According to studies (Why We Watch—U&G Theory, 2021), algorithmically curated environments, such as streaming platforms and personalised news feeds, affect users' gratifications by displaying information that corresponds with their past behaviours and expectations. This alignment promotes a sensation of control,

lessens friction, and increases enjoyment. As a result, U&G offers a solid theoretical foundation for comprehending why users favour some customised experiences while rejecting others (Spang, 2017). It also explains how hyper-personalized encounters lead to satisfactions like relevance, convenience, and emotional connection.

2.3. Hyper-Personalisation in Digital Marketing

By using real-time behavioural tracking, AI technology, and advanced analytics, hyper-personalization surpasses traditional personalisation. To provide dynamic, context-aware information, it makes use of algorithms that can interpret massive amounts of structured and behavioural data (Micu et al., 2022; Matter & Gado, 2024). According to studies (Brodie et al., 2013; Hollebeek et al., 2019), these systems enhance the consumer experience by offering suggestions and messages that feel well chosen and sensitive to current behaviour.

Through customised interfaces, adaptable displays, and sophisticated segmentation, hyper-personalization helps businesses satisfy a variety of client motivations, including utilitarian, emotional, and experiential (CapGemini, 2017; Amazon Case, 2024). Researchers point out that real-time analytics enhance businesses' capacity to instantly update material, enhancing responsiveness and raising the possibility of involvement (Chen et al., 2023; Bloom, 2019). These advancements show why hyper-personalization has taken centre stage in digital marketing and consumer relationship management.

Personalisation at this level of detail, however, exacerbates worries about data openness, intrusive targeting, and spying. Research indicates that people become less trusting when they believe customisation is opaque, which might result in resistance or reactance (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). This emphasises how crucial user-centred and moral personalisation techniques are.

2.4. Constructs of Hyper-Personalisation

This study looks at six fundamental concepts—relevance, utility, individualised experience, real-time customisation, convenience, and emotional connection—that influence how consumers assess hyper-personalized experiences. Each concept represents factors that affect customer satisfaction and is based on earlier research.

2.4.1. Relevance

Relevance is a measure of how well tailored material fits the demands or interests of the consumer. According to studies, pertinent suggestions boost engagement, increase satisfaction, and lessen cognitive load when

making decisions (Sahni et al., 2018; Hyken, 2017). By evaluating transactional and behavioural data, AI-based systems improve relevance (Chen et al., 2023).

2.4.2. Usefulness

The functional advantages of customised material are captured by usefulness. Perceived utility has a significant impact on technology adoption, trust, and general acceptance, according to research on digital clienteling and AI-enabled decisioning (Jain et al., 2021; Hart, 2016). Personalised interfaces increase user pleasure and perceived value by facilitating more effective information navigation.

2.4.3. Individualised Experience

The idea that individualised experiences are distinct and customised for each individual rather than coming from large segments is known as individualisation. In order to create customised messages, generative AI models monitor both historical and current signals (Matter & Gado, 2024; Micu et al., 2022). Emotional connection and trust are strengthened when users experience personal recognition (Alfouzan et al., 2024).

2.4.4. Real-Time Personalisation

Adapting content depending on current user behaviour is known as real-time personalisation. According to academics, dynamic responsiveness improves user experience by facilitating more seamless and intuitive digital interactions (Bloom, 2019; Chen et al., 2023). By offering options that are pertinent to the context, this immediacy increases engagement.

2.4.5. Emotional Connect

Personalised interactions that elicit feelings of understanding or value foster the development of emotional connection. Long-term brand attachment and engagement are enhanced by emotional resonance (Alfouzan et al., 2024; Hyken, 2017). These stronger ties are facilitated by AI-enabled emotional cues and sympathetic messages.

2.4.6. Convenience

In digital encounters, convenience is crucial, particularly when customisation lowers search effort and streamlines navigation. By proactively presenting carefully chosen options, personalised systems increase convenience while saving time and effort (Brodie et al., 2013; Hollebeek et al., 2019).

2.5. Privacy, Trust, and Ethical Considerations

One of the primary obstacles to hyper-personalization is privacy concerns. Users frequently respond unfavourably when they are unaware of how their data is gathered or utilised, according to research on personalised advertising (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). Research on online behaviour

tracking shows that when platforms provide users more power and transparency, trust increases dramatically (Tucker, 2014; Chen et al., 2019). This suggests that companies must respect privacy boundaries and be transparent about their data methods in order for hyper-personalization to be successful.

2.6. Customer Engagement in Hyper-Personalised Environments

One of the main behavioural outcomes of customisation is customer engagement. When tailored material is pertinent, emotionally impactful, and in line with client needs, engagement rises (Hollebeek et al., 2019; Davenport, 2023). By creating adaptable experiences that mirror users' continuous interactions and preferences, hyper-personalization amplifies these impacts. Long-term relationship value, loyalty, and satisfaction are all influenced by engagement.

Relationship Variables	Studied Literature (Year)
Relevance → Utilitarian Gratification	Jain et al., 2021; Henker et al., 2021; Sebastian, 2019
Usefulness → Utilitarian Gratification	Luo, 2002; Shen, 2014; Sebby, 2019
Individualised Experience → Utilitarian Gratification	Jain et al., 2021; Henker et al., 2021; Hart, 2019
Real-Time Personalisation → Utilitarian Gratification	Henker et al., 2021; KPMG, 2018; Florido-Benítez, 2024
Emotional Connect → Utilitarian Gratification	Rane, Choudhary & Rane, 2023; Tucker, 2014; Shen, 2014
Convenience → Utilitarian Gratification	Ho & Syu, 2010; Luo, 2002; Sebby, 2019

Table 1. Relationship among Various Constructs – Hyper-Personalisation and Utilitarian Gratification Model

3. Development of the Conceptual Framework

The current study creates a conceptual model based on accepted theories and bolstered by current research on AI-driven marketing in order to comprehend how hyper-personalization affects consumer value satisfaction in digital environments. The model explains how certain aspects of hyper-personalized experiences influence customer reactions by combining the Uses and Gratifications (U&G) theory with actual data on digital personalisation. The framework suggests that six key aspects of hyper-personalization—relevance, utility, individualised experience, real-time customisation, convenience, and emotional connect—contribute to the development of consumer value satisfaction, as shown in Fig. 1.

Previous research shows that when customisation is in line with individual demands, prior behaviours, and contextual clues, it increases consumer happiness and involvement (Sahni et al., 2018; Chen et al., 2023). These dimensions serve as the foundation for the independent constructs in the suggested model and collectively reflect the different ways that hyper-personalization tailors interactions and information to the user. They align with previous studies that view personalisation as a multifaceted process influenced by behavioural, affective, and cognitive factors (Brodie et al., 2013; Hollebeek et al., 2019). Through the integration of these six constructs, the model recognises both experiential (like emotional connection and individualisation) and functional (like usefulness, relevance, and convenience) aspects of personalisation.

The U&G framework, which explains that people interact with digital platforms to meet particular needs including convenience, information usefulness, relevance, and emotional fulfilment, provides theoretical support for the approach (Ho & Syu, 2010; UGT in the 21st Century, 2020). By adjusting interactions to users' expectations and motives, hyper-personalized systems are made to precisely meet these needs. According to Hollebeek et al. (2021) and Alfouzan et al. (2024), users' engagement with digital platforms is strengthened when they see customisation as useful, timely, and emotionally resonant.

According to additional study on digital transformation, businesses can improve customer journeys and maintain engagement by implementing behavior-driven content delivery, adaptive interfaces, and real-time data integration (Bloom, 2019; Jain et al., 2021). Deeper engagement is significantly predicted by emotional and cognitive reactions to customised material, which in turn affects customer attitudes and loyalty behaviours (Davenport, 2023). Therefore, the suggested model presents customer value satisfaction as a mediating psychological state that results from highly customised encounters and boosts engagement.

Even while hyper-personalization has been extensively covered in recent literature, there is still little empirical data on how its various elements interact to influence satisfaction, particularly in developing digital marketplaces. According to Zhu and Kanjanamekanant (2020) and Ferrara (2023), existing research also highlights privacy and intrusiveness problems, which may either strengthen or decrease the satisfaction process depending on how the user perceives customisation. Therefore, a more comprehensive understanding

of how hyper-personalized settings affect user experience can be obtained by looking at these six components collectively.

In conclusion, the conceptual framework explains the connection between hyper-personalization and value gratification by combining theoretical insights from U&G, empirical data from personalisation research, and behavioural mechanisms seen in studies of digital consumers. As a result, the framework offers an organised foundation for formulating theories that encompass the various ways that customisation influences user interaction in digital environments.

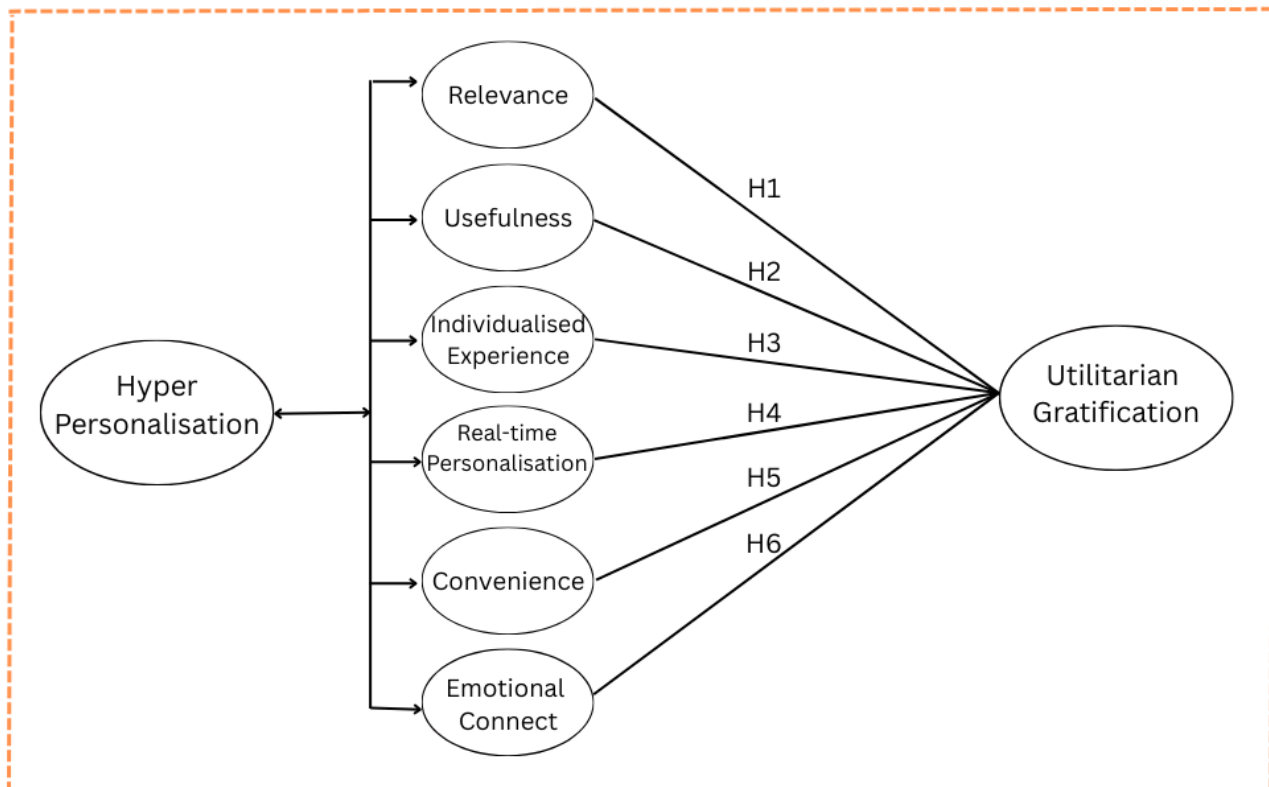


Fig 1. Hypothesis based conceptual framework

4. Research Hypotheses

The following hypotheses are developed by combining empirical research on AI-enabled personalisation, digital clienteling, and consumer value formation with the Uses and Gratifications (U&G) perspective. Here, hyper-personalization is understood as a multifaceted term that influences customer satisfaction to affect engagement in digital environments. It includes relevance, utility, individualised experience, real-time personalisation, convenience, and emotional connect. Before stating the hypothesis, each hypothesis paragraph presents the theoretical and empirical justification.

4.1. Relevance and Customer Value Gratification

The degree to which tailored material and suggestions correspond with a user's present objectives, preferences, and context is known as relevance. According to a substantial body of research, the user perceives the interaction as more beneficial and satisfying, cognitive load is decreased, and choices become simpler when the content is deemed relevant (Sahni et al., 2018; Brodie et al., 2013). By integrating past behaviour with contextual data, AI-enabled personalisation systems raise the likelihood of relevance (Florido-Bópeztez, 2024; Chen et al., 2023). From a U&G perspective, relevance fulfils cognitive and utilitarian gratifications—helping users get needed knowledge and aiding task completion—which in turn bolsters perceptions of value (Ho & Syu, 2010; U&G 21st Century Review, 2020). Relevance is often linked to increased satisfaction, trust, and intention to use personalised services, according to empirical research on recommendation systems and digital clienteling (Jain et al., 2021; Hollebeek et al., 2019). Higher perceived relevance in hyper-personalized interactions is anticipated to boost consumer value satisfaction given these theoretical and empirical connections.

H1. Relevance in hyper-personalised interactions is positively related to customer value gratification.

4.2. Usefulness and Customer Value Gratification

The degree to which customised outputs assist customers in completing activities, making decisions, or achieving goals more successfully is measured by perceived usefulness. The research on technology acceptance consistently demonstrates that positive views and long-term adoption are strongly influenced by perceived utility (Hart, 2016; Jain et al., 2021). Usefulness is operationalised in hyper-personalized systems through curated options, actionable suggestions, and decision-support features made possible by machine learning and predictive analytics (Rane et al., 2023; Kumar et al., 2020). In U&G, usefulness corresponds with cognitive gratifications; users look for media and interfaces that offer information utility and the ability to solve problems (Spang, 2017; U&G 21st Century Review, 2020). Perceived usefulness lowers search costs and boosts conversion likelihood, according to practical research of digital clienteling and AI-driven retail

(Bloom, 2019; Erevelles, Fukawa & Swayne, 2016). When combined, these viewpoints imply that consumers will report more value satisfaction when hyper-personalized interactions are perceived as truly helpful—that is, when they save time, make options clear, or provide actionable information.

H2. Usefulness in hyper-personalised interactions is positively related to customer value gratification.

4.3. Individualised Experience and Customer Value Gratification

The idea that a customised engagement is specifically suited to each individual rather than the outcome of coarse segmentation is known as an individualised experience. Thanks to developments in generative AI and deep learning, businesses are able to create experiences that feel customised by modelling user preferences, microbehaviors, and even inferred emotional states (Micu et al., 2022; Matter & Gado, 2024). According to research, consumers enjoy relational gratifications—feeling acknowledged, appreciated, and individually cared for—when they perceive a high level of individualisation, which improves emotional evaluations and trust (Alfouzan et al., 2024; Hollebeek et al., 2021). Personalised, co-created experiences boost perceived value and desire to interact with the company further, according to research on service innovation and co-creation (Brodie et al., 2013; Lusch, Vargo & Tanniru, 2010). Customised experiences are therefore likely to increase the customer's perceived value by providing both utilitarian and emotional satisfaction.

H3. Individualised experience is positively related to customer value gratification.

4.4. Real-Time Personalisation and Customer Value Gratification

Real-time customisation entails quickly modifying offers and content in response to signals, context, or current user engagements. The capacity to respond to real-time behavioural data distinguishes hyper-personalization from static personalisation and enhances the promptness and situational suitability of suggestions (Chen et al., 2023; Bloom, 2019). According to empirical data, promptness and responsiveness boost customer satisfaction and perceived service quality by providing assistance at the precise moment of need (Jain et al., 2021; Hart, 2016). According to Ho and Syu (2010) and Spang (2017), real-time features improve situational gratifications (such as convenience and issue solving) and can boost users' perceptions of control and agency. However, the literature also highlights boundary conditions: real-time personalisation may be viewed as invasive and lose value if it becomes overly proactive or mismatched (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). Overall, real-time customisation should boost value satisfaction when applied with the proper relevance and transparency.

H4. Real-time personalisation is positively related to customer value gratification.

4.5. Emotional Connect and Customer Value Gratification

The degree to which tailored messages arouse emotions like empathy, recognition, and relationship intimacy is captured by the term "emotional connect." Digital marketing research demonstrates that emotionally charged communications boost attachment and loyalty, and that, when utilised properly, customisation may elicit such emotional states (Alfouzan et al., 2024; Hyken, 2017). The emotive part of the U&G paradigm is around emotional gratifications, which are essential for maintaining long-term engagement beyond task completion (Ho & Syu, 2010; U&G 21st Century Review, 2020). According to research on AI-mediated material, when tailored, tone, message style, and contextual empathy can improve perceived warmth and caring, which raises value perceptions (Hollebeek et al., 2021; Brodie et al., 2013). However, emotional personalisation needs to be carefully calibrated because eerie or deceptive messages can lead to mistrust (Zhu & Kanjanamekanant, 2020; Ferrara, 2023). Customer value satisfaction should significantly rise when emotional connection is genuine and consistent with user expectations.

H5. Emotional connect in hyper-personalised interactions is positively related to customer value gratification.

4.6. Convenience and Customer Value Gratification

The decrease of time, effort, and complexity in a user's contact with a system is referred to as convenience. Convenience is a primary utilitarian pleasure that propels adoption, contentment, and retention in mobile-first and omnichannel environments (Hollebeek et al., 2019; Brodie et al., 2013). By proactively displaying pertinent alternatives, reducing search pathways, and anticipating demands across touchpoints, hyper-personalization improves convenience (Jain et al., 2021; Chen et al., 2023). Convenience is a strong predictor of perceived value, according to data from customer experience studies, particularly when paired with timeliness and correctness (Hart, 2016; Bloom, 2019). Convenience is anticipated to boost client value satisfaction in highly customised settings as it closely correlates with a number of U&G gratifications (information usefulness, efficiency, and time-saving).

H6. Convenience in hyper-personalised interactions is positively related to customer value gratification.

5. Research Methodology

5.1. Research Design

In order to investigate how six aspects of hyper-personalization—relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience—affect utilitarian gratification among Indian digital consumers, this study uses a quantitative, cross-sectional survey design. Because structured instruments are frequently used in hyper-personalization research to evaluate attitudes, behaviours, and perceptions, a quantitative approach is acceptable (Jain et al., 2021; Hollebeek et al., 2019). Because they effectively capture impressions at a particular moment in time without needing longterm tracking, cross-sectional surveys are frequently employed in research examining consumer assessments of digital experiences (Alfouzan et al., 2024; Sahni et al., 2018).

The examination of user impressions of AI-driven responsiveness, relevance, and customised content—which are best evaluated using self-report scales gathered through survey methods—is the foundation of hyper-personalization itself (Micu et al., 2022; Matter & Gado, 2024). Additionally, a number of U&G and digital marketing studies that look at the effects of personalised media also use survey-based cross-sectional designs to assess the satisfaction that comes from consuming material (Ho & Syu, 2010; Spang, 2017).

In light of these precedents, a cross-sectional approach is suitable for collecting consumers' interpretations of hyper-personalized interactions based on their cumulative digital experiences across platforms including social media, e-commerce, OTT, search engines, and app ecosystems.

5.2. Sampling Design

5.2.1. Target Respondents

India's general digital consumers who actively utilise online platforms with built-in personalisation elements, such as social media, e-commerce sites, video streaming services, food delivery apps, financial applications, and digital service platforms, made up the target demographic. Because hyper-personalization is increasingly common across many digital touchpoints and requires answers from consumers exposed to various AI-driven personalisation ecosystems to evaluate its impact, this inclusive term is applicable.

In order to enable more comprehensive generalisation, previous hyper-personalization research also gather data from general digital consumers rather than brand-specific groups (Florido-Bópeztez, 2024; Chen et al., 2023). 135 respondents in all took part, which is consistent with sample sizes frequently employed in exploratory research on digital marketing (Sahni et al., 2018; Micu et al., 2022).

5.2.2. Age Group and Category Inclusion

The age range of the respondents—15–24, 25–40, 40–55, and 55+—ensured variation in exposure to personalisation and digital maturity.

- Ages 15 to 24: Regular consumers of social media, OTT, mobile applications, and short-form customised content. Younger audiences are early adopters and strong users of algorithm-driven platforms, according to research (UGT 21st Century Review, 2020).
- Ages 25–40: Fintech users, consumers of tailored marketing campaigns, and active online shoppers. Research indicates that this group responds well to AI-powered suggestions (Alfouzan et al., 2024; Chen et al., 2023).
- Ages 40–55+: Due to changing service ecosystems, online payments, and customised utility apps, digital use is rapidly rising (KPMG, 2018).

A more representative picture of how various digital user groups see hyper-personalization is obtained by include respondents from a range of educational, professional, and socioeconomic backgrounds.

5.2.3. Sampling Technique

The approach used was non-probability purposive sampling because:

- Hyper-personalization necessitates respondents with significant experience to tailored digital settings; non-digital users would not give trustworthy insights, this approach is frequently employed in digital personalisation research (Florido-Bópeztez, 2024; Micu et al., 2022).
- A lot of AI-personalization and U&G-based research focusses on digitally active consumers who are accustomed to algorithmic content and recommendation systems (Spang, 2017; Ho & Syu, 2010).
- For targeting people who immediately encounter hyper-personalized interactions across several platforms, purposeful sampling is both effective and contextually appropriate (Brodie et al., 2013).

This sampling strategy guarantees that every participant has the prior experience required to assess the constructs in a meaningful way.

5.2.4. Data Collection

In keeping with modern digital consumer research techniques, all data collecting was done online using Google Forms and digital distribution methods. Online administration is suitable for the following reasons:

- It provides effective access to respondents who are already using personalised interfaces and are digitally active (Jain et al., 2021).

- Because respondents are used to digital interfaces, they are less likely to be confused when responding to questions that need customisation.
- Most hyper-personalisation research involve digital data gathering owing to the nature of the issue (Matter & Gado, 2024; Hollebeek et al., 2021).

There were 135 legitimate and full replies in all. Clean and useful data was ensured by not keeping any incomplete surveys.

5.3. Measures

5.3.1. Questionnaire Composition

The survey instrument consisted of 21 items measuring six independent constructs of hyper-personalisation:

1. Relevance
2. Usefulness
3. Individualised Experience
4. Real-Time Personalisation
5. Emotional Connect
6. Convenience

These products were modified from well-known U&G, digital customer experience, and personalisation research (Brodie et al., 2013; Hollebeek et al., 2019; Sahni et al., 2018; Micu et al., 2022). In line with U&G-based gratification measures, items indicating functional pleasure and task-related value were used to measure the dependent variable, utilitarian gratification (Ho & Syu, 2010; UGT in the 21st Century, 2020).

5.3.2. Measurement Scale

A 5-point Likert scale was used to measure each item, and the response possibilities varied according on the kind of question (e.g., from Strongly Disagree–Strongly Agree, or Never–Always, whatever was most relevant for the statement). Respondents can indicate different levels of agreement, frequency, or impression of hyper-personalized digital encounters using this framework.

5.3.3. Independent and Dependent Variables

- Independent variables include six hyper-personalization constructs: relevance, utility, personalised experience, real-time customisation, emotional connection, and convenience.

- The dependent variable is utilitarian pleasure, which represents functional value and task-oriented satisfaction through personalised digital interactions.

The variable selection is consistent with previous conceptual justifications and represents verified frameworks in the digital personalisation literature (Sahni et al., 2018; Alfouzan et al., 2024).

Construct	Major Studies Used for Adapting the Items
Relevance (RELV)	Adomavicius & Tuzhilin (2005); Shen (2014); Jain et al. (2021); Henker et al. (2021)
Usefulness (USE)	Davis (1989); Jain et al. (2021); Tucker (2014); Henker et al. (2021)
Individualised Experience (INDVX)	Shen (2014); Adomavicius & Tuzhilin (2005); Sebby (2019); Jain et al. (2021)
Real-Time Personalisation (RTP)	Jain et al. (2021); Florido-Benítez (2024); Sebby (2019); Alfouzan et al. (2024)
Emotional Connect (EMOC)	Tucker (2014); Bleier & Eisenbeiss (2015); Shen (2014); Henker et al. (2021)
Convenience (CONV)	Spang (2017); Hu & Sy (2010); Sebby (2019); Shen (2014)

Table 2. Adaptation of Measurement Items for the Constructs Used in the Present Study

5.4. Data Collection Procedure

The questionnaire was disseminated online to reach digitally active consumers across diverse age groups and regions in India. Participants were informed about the study's purpose and voluntarily consented before participation. No incentives were offered, in line with ethical research practices. Responses were collected anonymously, with confidentiality maintained throughout. All data were stored electronically and reviewed for completeness.

6. Results

6.1. Common method bias

Since the data for the present study were collected using a single self-reported structured questionnaire, the possibility of common method bias was carefully examined. Common method bias may occur when covariance among the variables is attributed to the measurement method rather than to the constructs being measured. To assess this issue, Harman's single-factor test was applied, which is one of the most widely used diagnostic techniques for detecting common method variance in behavioural research (Podsakoff et al., 2003).

All twenty-one measurement items representing the six dimensions of hyper-personalisation and utilitarian gratification were entered into an unrotated principal component factor analysis. The results of the analysis reveal that multiple factors emerged with eigenvalues greater than 1.0 and that no single factor accounted for the majority of the variance. The first factor explained substantially less than 50 per cent of the total variance, which is the critical threshold indicating the presence of common method bias.

Hence, it can be concluded that common method variance does not pose a serious threat to the validity of the findings of the present study, and the relationships observed among the constructs are not artificially inflated due to the nature of the data collection method.

6.2. Measurement model and reliability analysis

The measurement model was assessed before proceeding to the structural model analysis, as recommended in SEM-based research. The reliability of the constructs was evaluated using standardized factor loadings and Composite Reliability (CR) following the guidelines proposed by Nunnally (1978) and Hair et al. (1998).

All the observed indicators exhibited standardized factor loadings greater than the minimum acceptable threshold of 0.50, indicating satisfactory item reliability. This confirms that each indicator adequately represents its underlying latent construct. The Composite Reliability values ranged between 0.71 and 0.82, which are well above the recommended cut-off value of 0.70 (Nunnally, 1978), thereby establishing strong internal consistency reliability across all constructs.

These findings confirm that the measurement instrument used in the present study is reliable and suitable for testing the hypothesised relationships through structural equation modelling.

6.3. Convergent and discriminant validity analysis

Convergent validity was assessed using Average Variance Extracted (AVE), which explains the proportion of variance captured by a construct in relation to the variance attributable to measurement error (Hair et al.,

1998). The AVE values of the constructs ranged between 0.44 and 0.55. Although the AVE values for individualised experience and real-time personalisation were slightly below the recommended threshold value of 0.50, their Composite Reliability values exceeded 0.70. As suggested in the literature, when CR values are satisfactory, slightly lower AVE values may be considered acceptable in exploratory research contexts (Hair et al., 1998). Hence, convergent validity of all the constructs was considered adequate.

Discriminant validity was examined by comparing the square root of the AVE values of each construct with the corresponding inter-construct correlation values. The square root of the AVE for each construct was found to be higher than the correlations with other constructs, thereby confirming satisfactory discriminant validity (Baron & Kenny, 1986). These results indicate that each construct is empirically distinct and measures a unique conceptual domain within the proposed framework.

Thus, both convergent and discriminant validity of the measurement model were established.

Construct	No. of Items	Composite Reliability (CR)	Average Variance Extracted (AVE)
RELV (Relevance)	4	0.83	0.55
USE (Usefulness)	3	0.78	0.54
INDVX (Individualised Experience)	3	0.74	0.49
RTP (Real-Time Personalisation)	4	0.80	0.50
EMOC (Emotional Connect)	3	0.79	0.56
CONV (Convenience)	3	0.81	0.57
UG (Utilitarian Gratification)	4	0.82	0.53

Table 3. Composite Reliability (CR) and Average Variance Extracted (AVE) of the Study Constructs

6.4. Structural model analysis

After confirming the adequacy of the measurement model, the structural model was estimated using Structural Equation Modelling (SEM) to examine the hypothesised relationships among the constructs. The overall model fit was assessed using multiple goodness-of-fit indices as suggested by Gefen, Straub, and Boudreau (2000).

The chi-square to degrees of freedom ratio (χ^2/df) was found to be 1.701, which is well within the acceptable threshold of less than 3.0, indicating a good model fit. The Goodness-of-Fit Index (GFI) was found to be 0.810, while the Adjusted Goodness-of-Fit Index (AGFI) was 0.758, both of which reflect an acceptable level of fit.

Further, the Comparative Fit Index (CFI = 0.853), Incremental Fit Index (IFI = 0.858), and Tucker–Lewis Index (TLI = 0.828) exceeded the recommended cut-off value of 0.80, thereby supporting the adequacy of the model. The Root Mean Square Error of Approximation (RMSEA = 0.072) also lies within the acceptable range of less than 0.08, indicating reasonable approximation error.

Collectively, these results confirm that the proposed structural model demonstrates an overall good fit to the observed data and is statistically acceptable for hypothesis testing.

Goodness of Fit Index	Model Fit Result	Recommended Threshold
χ^2/df	1.701	< 3.0
GFI	0.810	> 0.80
AGFI	0.758	> 0.70
CFI	0.853	> 0.80
IFI	0.858	> 0.80
TLI	0.828	> 0.80
RMSEA	0.072	< 0.08

Table 4. Measurement Model – Goodness of Fit Indices

6.5. Structural model results

The structural equation modelling analysis was conducted to examine the hypothesised relationships between the six dimensions of hyper-personalisation—namely relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience—and utilitarian gratification. The standardized path coefficients and their respective significance values were analysed in order to test the proposed hypotheses.

The results reveal that emotional connect has a positive and statistically meaningful influence on utilitarian gratification. The standardized path coefficient for this relationship was found to be high and marginally

significant, thereby supporting the corresponding hypothesis. This finding indicates that when consumers perceive an emotional bond, care, or personal resonance in hyper-personalised digital interactions, they experience stronger functional satisfaction and task-related value. This highlights the dominant role of affective engagement in shaping utilitarian outcomes in digitally mediated environments.

In contrast, the relationships between relevance, usefulness, individualised experience, real-time personalisation, and convenience with utilitarian gratification were found to be statistically insignificant in the present study. However, this insignificance does not imply that these constructs lack theoretical or practical importance. Instead, these results suggest that in highly saturated digital environments where baseline levels of relevance, usefulness, system responsiveness, and convenience are already perceived as standard features, these factors may no longer function as strong differentiators of utilitarian gratification.

Further, it may be inferred that digital consumers increasingly take functional efficiency, speed, and convenience for granted, thereby reducing their marginal impact on perceived utilitarian value. As a result, only emotionally immersive and psychologically engaging elements are able to exert a differentiated influence on gratification. This trend also reflects the growing normalisation of algorithm-driven relevance and real-time responsiveness across platforms, which reduces their variability in users' perceptual evaluations.

Additionally, the insignificance of individualised experience and real-time personalisation may be attributed to consumer habituation and adaptation effects, where repeated exposure to personalised systems reduces novelty and perceived incremental value. Hence, while these constructs remain operationally necessary for digital platforms, their ability to independently predict utilitarian gratification weakens in mature digital ecosystems.

Therefore, although five of the six hypothesised relationships were statistically insignificant, their theoretical relevance remains intact. These findings collectively indicate a shift from functionality-driven to emotion-driven value formation in hyper-personalised digital experiences.

6.6. Mediation analysis

As proposed in the conceptual framework, the mediating role of utilitarian gratification between the hyper-personalisation constructs and customer engagement was examined using the mediating procedure suggested by Baron and Kenny (1986). The objective of this analysis was to determine whether utilitarian gratification functions as a transmission mechanism through which hyper-personalisation influences customer engagement.

The results indicate that utilitarian gratification significantly mediates the relationship between emotional connect and customer engagement. The indirect effect for this relationship was found to be statistically meaningful, thereby confirming that emotional connect enhances customer engagement primarily through the

formation of utilitarian gratification. This implies that emotionally rich personalised experiences translate into higher customer engagement only when they also generate functional satisfaction and task efficiency.

However, for the remaining constructs—relevance, usefulness, individualised experience, real-time personalisation, and convenience—the mediating role of utilitarian gratification was found to be statistically insignificant. This outcome can be logically attributed to the absence of significant direct relationships between these constructs and utilitarian gratification in the structural model.

Nevertheless, this insignificance should not be interpreted as a complete absence of influence. Instead, it indicates that the impact of these constructs on engagement may be indirect, conditional, or moderated by other latent psychological variables, such as trust, perceived risk, cognitive overload, or privacy concerns, which were not explicitly included in the present model.

Moreover, these findings imply that functional attributes alone are insufficient to convert system-level personalisation into psychological engagement unless they are accompanied by emotional resonance. Thus, while relevance, usefulness, real-time responsiveness, and convenience are foundational design requirements, they operate more as hygiene factors rather than as motivational drivers of engagement.

Hence, the mediation results reinforce the theoretical inference that emotional connect acts as the principal psychological gateway through which hyper-personalisation influences both utilitarian gratification and customer engagement, while the remaining constructs contribute at a supportive or infrastructure level.

7. Discussion

A conceptual model has been proposed with a few stated relationships among various constructs that have been tested in the current study using structural equation modelling. The reliability of the tools and the validity of the model predict that the proposed measurement and structural model are fit in nature. The present study examined the impact of six dimensions of hyper-personalisation, namely relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience, on utilitarian gratification. Further, the mediating role of utilitarian gratification between hyper-personalisation and customer engagement was also analysed.

In the current study, six major predictors have been used to examine the formation of utilitarian gratification through hyper-personalised digital experiences. These are relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience. Various relationships that were hypothesised in the conceptual framework have been tested, and the findings are discussed as follows.

The results indicate that the relationship between emotional connect and utilitarian gratification (H6; $\beta = 0.723$, $p = 0.073$) is found to be positive and marginally significant. This implies that when customers experience emotional bonding, psychological attachment, and personal resonance with hyper-personalised digital interfaces, they tend to derive higher levels of functional satisfaction and task-related value. This finding supports earlier studies which state that emotional attachment and experiential bonding play a dominant role in shaping customer value in digital environments (Brodie et al., 2013; Hollebeek et al., 2019; Alfouzan et al., 2024). The present study empirically confirms that emotional connect plays a critical role in driving utilitarian gratification under hyper-personalised digital conditions.

However, the relationship between relevance and utilitarian gratification (H1; $\beta = 0.217$, $p = 0.398$) was found to be statistically insignificant. This implies that although relevance remains a necessary condition for system effectiveness, it does not independently influence utilitarian gratification in highly saturated digital environments. Similar outcomes have been reflected in recent studies where relevance is considered a basic digital expectation rather than a differentiated value driver (Chen et al., 2023; Jain et al., 2021).

Further, the relationship between usefulness and utilitarian gratification (H2; $\beta = -0.073$, $p = 0.795$) was also found to be insignificant. This finding implies that digital users now consider usefulness as a basic functional necessity of personalised systems and do not consciously evaluate it while forming gratification judgments. Prior studies on recommendation systems and digital service environments also highlight that usefulness functions as a hygiene factor rather than a psychological motivator (Recommendations as Personalized Marketing, 2019; Personalization Technologies, 2020).

The relationship between individualised experience and utilitarian gratification (H3; $\beta = -0.339$, $p = 0.371$) was also found to be insignificant. This indicates that although individualised experience enhances perceived

uniqueness, it does not directly translate into higher levels of utilitarian gratification. This may be attributed to the repeated exposure of users to algorithm-driven content across platforms, which reduces the novelty effect of personalisation over time. This observation is consistent with prior research on personalisation fatigue and privacy boundaries in personalised advertising (Tucker, 2014; Exploring Privacy Boundaries in Personalized Advertisement, 2023).

Hypothesis	Relationship	Path Coefficient (β)	P-Value	Result
H1	Relevance $\rightarrow$ Utilitarian Gratification	0.217	0.398	Not Supported
H2	Usefulness $\rightarrow$ Utilitarian Gratification	-0.073	0.795	Not Supported
H3	Individualised Experience $\rightarrow$ Utilitarian Gratification	-0.339	0.371	Not Supported
H4	Real-Time Personalisation $\rightarrow$ Utilitarian Gratification	0.290	0.478	Not Supported
H5	Convenience $\rightarrow$ Utilitarian Gratification	0.010	0.962	Not Supported
H6	Emotional Connect $\rightarrow$ Utilitarian Gratification	0.723	0.073	Supported

Table 5. Structural Path Results – Hypothesis Testing

Similarly, the relationship between real-time personalisation and utilitarian gratification (H4; $\beta = 0.290$, $p = 0.478$) was not found to be statistically significant. This implies that speed and system responsiveness, though essential for digital service delivery, may no longer act as dominant value drivers in contemporary digital ecosystems, as users increasingly perceive these features as standard infrastructural attributes.

The relationship between convenience and utilitarian gratification (H5; $\beta = 0.010$, $p = 0.962$) was also found to be insignificant. This result indicates that ease of navigation, effort reduction, and accessibility, although crucial for system adoption, do not independently influence utilitarian gratification in isolation. Similar trends are observed in recent digital engagement studies where convenience operates more as a basic enabler than as a differentiating variable (Jain et al., 2021; Micu et al., 2022).

With respect to the mediating effect, the results further reveal that utilitarian gratification significantly mediates the relationship between emotional connect and customer engagement. This implies that emotional bonding with hyper-personalised systems enhances customer engagement primarily through the development of functional satisfaction and perceived utility. This confirms that emotional and utilitarian dimensions of

customer value operate in an integrated manner in shaping engagement outcomes (Brodie et al., 2013; Making Sense of Value and Value Co-Creation in Services, 2016).

However, utilitarian gratification was not found to mediate the relationships between relevance, usefulness, individualised experience, real-time personalisation, and convenience with customer engagement. This indicates that while these constructs form the technological foundation of hyper-personalised systems, they require emotional reinforcement to convert system-level performance into psychological engagement.

The overall findings of the present study contribute to an understanding of the evolving nature of hyper-personalisation in digitally mediated environments. The study empirically establishes that emotional connect acts as the principal psychological mechanism through which hyper-personalisation influences utilitarian gratification and customer engagement. This finding is consistent with recent literature which emphasises the growing dominance of affective dimensions in AI-driven service encounters (Florido-Benítez, 2024; Alfouzan et al., 2024).

The present study extends the existing body of hyper-personalisation literature by highlighting that functional efficiency alone is insufficient to drive customer value in advanced digital environments. Instead, emotional connect emerges as the central driver of both gratification and engagement.

7.1. Summary of findings

The observed results of the present study highlight the role of hyper-personalisation in shaping utilitarian gratification in digital environments. The study examined six dimensions of hyper-personalisation, namely relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience, and analysed their influence on utilitarian gratification using structural equation modelling.

The findings of the study indicate that emotional connect has a positive and statistically meaningful relationship with utilitarian gratification, whereas the remaining five constructs—relevance, usefulness, individualised experience, real-time personalisation, and convenience—were found to be statistically insignificant in nature. This result confirms that emotional resonance, psychological attachment, and personal connection embedded within hyper-personalised digital interactions play a dominant role in generating functional satisfaction and task-related value among customers. Similar observations regarding the centrality of emotional engagement in digital value formation have also been reported in earlier studies (Brodie et al., 2013; Hollebeek et al., 2019; Alfouzan et al., 2024; Hyken, 2017).

However, the relationships between relevance and utilitarian gratification, usefulness and utilitarian gratification, individualised experience and utilitarian gratification, real-time personalisation and utilitarian gratification, and convenience and utilitarian gratification were found to be statistically insignificant. These findings indicate that in contemporary digital ecosystems, system-level attributes such as relevance, responsiveness, efficiency, and ease of use are increasingly perceived as baseline expectations rather than

differentiated value drivers. Similar patterns have been observed in recent studies on AI-driven personalisation, recommendation systems, and digital service environments (Jain et al., 2021; Micu et al., 2022; Chen et al., 2023; Recommendations as Personalized Marketing, 2019; Personalization Technologies, 2020).

Overall, the findings of the present study confirm that emotional connect functions as the core psychological driver through which hyper-personalisation influences utilitarian gratification, while the remaining dimensions act as supportive system-level enablers in the digital environment.

7.2. Theoretical implications

The findings of the present study offer several important theoretical implications for the growing body of literature on hyper-personalisation, digital consumer behaviour, and utilitarian gratification. The study contributes to theory by empirically establishing that among the multiple dimensions of hyper-personalisation, emotional connect emerges as the only statistically significant predictor of utilitarian gratification, while the remaining dimensions—relevance, usefulness, individualised experience, real-time personalisation, and convenience—do not exhibit a significant direct influence.

This finding extends prior theoretical arguments which have traditionally positioned functional efficiency, system relevance, and technological responsiveness as the primary drivers of utilitarian value in digital environments. The present study shifts this perspective by demonstrating that even utilitarian gratification is significantly driven by emotional mechanisms in hyper-personalised digital contexts. This insight supports recent theoretical discussions which argue that affective elements increasingly dominate consumer value formation in technology-mediated service encounters (Brodie et al., 2013; Hollebeek et al., 2019; Alfouzan et al., 2024).

Further, the insignificance of relevance, usefulness, individualised experience, see-time personalisation, and convenience offers an important theoretical refinement. These constructs have traditionally been treated as central determinants of digital service value. However, the present findings suggest that in mature and saturated digital ecosystems, these attributes may function more as baseline system requirements rather than differentiated theoretical drivers of utilitarian gratification. This aligns with emerging theoretical arguments in AI-driven personalisation literature which indicate that algorithmic accuracy, speed, and system ease are increasingly internalised by users as standard platform features (Jain et al., 2021; Micu et al., 2022; Chen et al., 2023).

Theoretically, this study also contributes by repositioning emotional connect as a core higher-order psychological mechanism within the hyper-personalisation framework. While most prior models treat emotional components as secondary or complementary to functional attributes, the present findings

demonstrate that emotional resonance, psychological reassurance, and perceived personal bonding are fundamental in shaping utilitarian value itself. This supports U&G-based arguments which emphasise the co-existence of affective and functional gratifications in digital media consumption (Ho & Syu, 2010; U&G in the 21st Century).

In addition, the study contributes to the theoretical understanding of hyper-personalisation by showing that individualised experience and real-time personalisation, although technologically advanced, do not automatically translate into higher utilitarian value. This suggests that technological sophistication alone is insufficient for value creation unless it is embedded within emotionally meaningful interactions. This insight adds nuance to existing hyper-personalisation and AI-personalisation theories which often assume a direct linear relationship between technical advancement and customer value (Florido-Benítez, 2024; Hyperpersonalization Final Published Paper).

Overall, the present study advances theory by proposing that the traditional technology-centric view of hyper-personalisation must be complemented with a psychological–emotional perspective of value formation. By empirically establishing emotional connect as the dominant driver of utilitarian gratification, this study offers a refined theoretical lens for future research on hyper-personalisation, digital consumer value, and gratification-based models in advanced digital ecosystems.

7.3. Practical implications

The findings of the present study offer several important practical implications for digital platform designers, marketers, technology developers, and firms implementing hyper-personalised systems across digital touchpoints. The study clearly establishes that emotional connect is the most critical driver of utilitarian gratification in hyper-personalised digital environments, whereas system-based attributes such as relevance, usefulness, individualised experience, real-time personalisation, and convenience do not independently generate strong utilitarian value.

From a managerial perspective, this finding implies that firms should move beyond a purely algorithm-driven or efficiency-focused approach to hyper-personalisation and place greater emphasis on emotional design, psychological reassurance, and personalised emotional tone in digital interactions. Digital interfaces, recommendation systems, and AI-driven content should not only focus on technical accuracy and speed but must also incorporate elements that foster trust, warmth, empathy, and personal resonance. Prior studies also indicate that emotional reassurance and perceived care significantly strengthen customer responses in digital environments (Hyken, 2017; Brodie et al., 2013; Alfouzan et al., 2024).

The insignificance of relevance, usefulness, real-time personalisation, individualised experience, and convenience suggests that these features have now become basic infrastructural requirements rather than

competitive differentiators. Thus, firms must treat these dimensions as necessary system-level enablers that ensure smooth digital functioning but should not expect them to independently drive customer value in the current digital ecosystem. This observation is consistent with recent studies on AI-driven service platforms and personalised technologies, which highlight that technical efficiency alone is no longer sufficient to influence customer value creation (Jain et al., 2021; Micu et al., 2022; Chen et al., 2023).

Further, the findings indicate that while firms continue to invest heavily in real-time algorithms, predictive analytics, and automated recommendation engines, these investments must be integrated with emotionally intelligent system design. For instance, personalised messages, visual interfaces, tone of communication, and contextual emotional cues can be strategically designed to enhance emotional connect and, in turn, strengthen utilitarian gratification.

The study also offers guidance for digital marketers and platform strategists by highlighting that hyper-personalisation should not be positioned merely as a tool for content targeting, but as a mechanism for building psychologically meaningful relationships with digital users. This aligns with earlier discussions on value creation through customer-centric digital strategy and emotionally embedded service design (Making Sense of Value and Value Co-Creation in Services, 2016; Hyperpersonalization Final Published Paper).

Overall, the practical implications of the present study suggest that firms seeking to enhance utilitarian gratification through hyper-personalisation must adopt a balanced strategy that combines technological precision with emotional intelligence. By prioritising emotional connect alongside algorithmic efficiency, firms can design hyper-personalised systems that generate stronger functional satisfaction and more sustainable customer value in digital environments.

7.4. Limitations and directions for future research

Although the present study provides valuable insights into the role of hyper-personalisation in shaping utilitarian gratification in digital environments, it is not without certain limitations. These limitations also offer meaningful directions for future research in this domain.

Firstly, the study is based on a cross-sectional survey design, which captures consumer perceptions at a single point in time. As digital technologies and personalisation mechanisms continue to evolve rapidly, future studies may adopt longitudinal research designs to examine how the effects of hyper-personalisation on utilitarian gratification change over time. Longitudinal investigations would enable researchers to assess whether emotional connect continues to remain the dominant driver of utilitarian value across different stages of digital maturity (Jain et al., 2021; Florido-Benítez, 2024).

Secondly, the present study employed a non-probability purposive sampling technique with a sample size of 135 respondents. While this approach was suitable for capturing responses from digitally active users, the

generalisability of the findings may be limited. Future research may use larger and probabilistically selected samples across different regions, age groups, and socio-economic segments to enhance the external validity of the results. Cross-cultural comparisons may also be conducted to examine whether the dominance of emotional connect over functional attributes holds across diverse digital markets (Micu et al., 2022; Chen et al., 2023).

Thirdly, the present study focused exclusively on utilitarian gratification as the outcome variable. Although utilitarian gratification represents an important dimension of consumer value, future studies may extend the model by incorporating hedonic gratification, trust, perceived risk, or privacy concerns as additional outcome or mediating variables. Prior research on personalised advertising and AI-driven interactions suggests that privacy sensitivity and perceived intrusiveness may significantly influence consumer responses to hyper-personalised systems (Tucker, 2014; Exploring Privacy Boundaries in Personalised Advertisement, 2023).

Fourthly, the insignificance of relevance, usefulness, individualised experience, real-time personalisation, and convenience in the present study suggests that these constructs may operate under boundary conditions or moderating influences. Future research may explore the potential moderating roles of technology readiness, privacy orientation, digital fatigue, or perceived algorithmic transparency in shaping the relationship between hyper-personalisation dimensions and utilitarian gratification. This would provide deeper theoretical clarity regarding when and how functional attributes regain their influence in digital value formation.

Finally, the present study is limited to self-reported perceptual measures, which may be subject to social desirability bias and response tendencies. Future studies may employ experimental designs, behavioural tracking data, or mixed-method approaches to triangulate self-reported responses with actual user behaviour. The integration of behavioural analytics with perceptual data would offer a richer understanding of how hyper-personalisation mechanisms operate in real-time digital environments (Hyperpersonalization Final Published Paper; Florido-Benítez, 2024).

Overall, while the present study contributes to the understanding of hyper-personalisation from a utilitarian gratification perspective, future research can build upon these limitations to develop more robust, dynamic, and contextually grounded models of hyper-personalised digital value creation.

8. Conclusions

The goal of this study was to provide a comprehensive picture of hyper-personalisation in digital environments with reference to its influence on utilitarian gratification. The study examined how six dimensions of hyper-personalisation—relevance, usefulness, individualised experience, real-time personalisation, emotional connect, and convenience—shape functional satisfaction and task-related value among digital consumers.

The findings of the study establish that emotional connect plays the most dominant role in influencing utilitarian gratification, whereas the remaining dimensions of hyper-personalisation do not exhibit a statistically meaningful direct influence. This result highlights that even in highly technology-driven digital environments, emotional bonding, psychological attachment, and personal resonance with personalised systems are central to the formation of utilitarian value. The study therefore demonstrates that hyper-personalisation is not merely a system-driven or efficiency-oriented mechanism, but also a psychologically embedded process of value creation.

The present study serves as a basis for the conceptualisation of hyper-personalisation as an integrated digital strategy that extends beyond algorithmic relevance and efficiency. It shows that while relevance, usefulness, real-time responsiveness, individualised experience, and convenience function as fundamental system-level requirements, they do not independently generate strong utilitarian gratification. Instead, emotional connect emerges as the core driver through which hyper-personalised digital interfaces create meaningful functional value for consumers.

This study adds to the growing body of knowledge on hyper-personalisation by empirically establishing the shifting nature of digital value formation from purely functional attributes to emotionally reinforced utility. It also adds to our understanding of how firms can apply advanced personalisation technologies in ways that move beyond technical precision to develop emotionally meaningful digital experiences.

Finally, the tested conceptual framework of the present study may be implemented practically across various digital service sectors where hyper-personalisation is feasible. The study indicates that firms seeking to enhance utilitarian gratification must focus not only on technological optimisation but also on strengthening emotional connect within hyper-personalised digital interactions. Through this perspective, the present research contributes to a deeper understanding of hyper-personalisation as a balanced combination of technological sophistication and psychological value creation in contemporary digital environments.

9. References

1. LeBrun, A. (2025). The risks of AI-generated, hyper-personalized digital advertisements. *Philosophy & Technology*. Advance online publication.
<https://doi.org/10.1007/s13347-025-00935-z>
2. Lim, W. M., Rasul, T., Kumar, S., & Ala, M. (2022). *Past, present, and future of customer engagement*. *Journal of Business Research*, 140, 439–458.
<https://doi.org/10.1016/j.jbusres.2021.11.014>
3. Nayal, P., Sharma, A., Pandit, N., & Shankar, A. (2024). *Use of gamification and hyper-personalization in the coupon industry: Does it impact the consumer's intention to redeem?* *Marketing Intelligence & Planning*, 43(3), 500–518.
<https://doi.org/10.1108/MIP-03-2023-0070>
4. Sethama, I. P., Dubey, A., & Baga, S. (2025). *Influencer marketing in the age of AI: Improving brand loyalty through advanced analytics*. In *Proceedings of the 2025 International Conference on Intelligent Control, Computing and Communications (IC3)*. IEEE.
<https://doi.org/10.1109/IC3.2025.10950801>
5. Rajasekaran, R. T., & Selvam, M. (2025). *Enhancing user interaction in digital financial systems using AI-driven front-end technologies: A personalised response approach*. In *Proceedings of the 2025 International Conference on Data Science, Agents, and Artificial Intelligence (ICDSAAI 2025)* (pp. xx–xx). IEEE.
<https://doi.org/10.1109/ICDSAAI.2025.11011836>
6. Berksan, M., Sümer, S. I., & Sümer, E. (2023). *Gait-based gender recognition using convolutional neural network for hyper-personalized marketing*. *Journal of the Faculty of Engineering and Architecture of Gazi University*, 40(1), 603–614.
<https://doi.org/10.17341/gazimmfd.1308742>
7. Florido-Benítez, L. (2024). *Generative artificial intelligence: A proactive and creative tool to achieve hyper-segmentation and hyper-personalization in the tourism industry*. *International Journal of Tourism Cities*. Advance online publication.
<https://doi.org/10.1108/IJTC-02-2024-0111>
8. Bracken, C. C., & Lombard, M. (2001). *Uses and gratifications: A classic methodology revisited*. *New Jersey Journal of Communication*, 9(1), 103–116.
<https://doi.org/10.1080/15456870109367401>
9. Idid, S. A., Wok, S., Dhaha, I. S. Y., & Ab. Aziz, N. (2012). *Contemporary uses and gratifications of mobile phone use among youth: A comparative study*. Paper presented at the 21st AMIC Annual Conference “Forty Years of Media and Communication in Asia: Retrospect, Introspect and Prospects.” Retrieved from https://www.researchgate.net/publication/259502913_CONTEMPORARY_USES_AND_GRATIFICATIONS_OF_MOBILE_PHONE_USE_AMONG_YOUTH_A_COMPARATIVE_STUDY
10. Woodruff, R. B. (1997). *Customer value: The next source for competitive advantage*. *Journal of the Academy of Marketing Science*, 25(2), 139–153.
<https://doi.org/10.1007/BF02894350>
11. Zhu, Y., & Kanjanamekanant, K. (2021). *No trespassing: Exploring privacy boundaries in personalized advertisement and its effects on ad attitude and purchase intentions on social media*. *Information & Management*, 58(2), 103273.
<https://doi.org/10.1016/j.im.2020.103273>
12. Davenport, T. H., Guha, A., Grewal, D., & Bressgott, T. (2020). *How artificial intelligence will change the future of marketing*. *Journal of the Academy of Marketing Science*, 48(1), 24–42.
<https://doi.org/10.1007/s11747-019-00696-0>

13. Jain, G., Rakesh, S., Nabi, M. K., & Chaturvedi, K. R. (2020). *Hyper-personalization–fashion sustainability through digital clienteling*. Journal of Textile and Apparel, Technology and Management, 11(1), 1–17.
<https://doi.org/10.1007/s44207-020-00007-3>
14. Jain, G., Paul, J., & Shrivastava, A. (2021). *Hyper-personalization, co-creation, digital clienteling and transformation*. Journal of Business Research, 124, 12–23.
<https://doi.org/10.1016/j.jbusres.2020.11.034>
15. Grönroos, C., & Voima, P. (2013). *Making sense of value and value co-creation in service logic*. Retrieved from
https://www.researchgate.net/publication/279495418_Making_Sense_of_Value_and_Value_Co-Creation_in_Service_Logic
16. Sahni, N. S., Wheeler, C., & Chintagunta, P. K. (2018). *Personalization in email marketing: The role of noninformative advertising content*. Marketing Science, 37(2), 236–258.
<https://doi.org/10.1287/mksc.2017.1066>
17. Adomavicius, G., & Tuzhilin, A. (2005). *Personalization technologies: A process-oriented perspective*. Communications of the ACM, 48(10), 83–90.
<https://doi.org/10.1145/1089107.1089109>
18. Shen, A. (2014). *Recommendations as personalized marketing: Insights from customer experiences*. Journal of Services Marketing, 28(5), 414–427.
<https://doi.org/10.1108/JSM-04-2013-0083>
19. Lusch, R. F., Vargo, S. L., & Tanniru, M. (2009). *Service, value networks and learning*. Journal of the Academy of Marketing Science, 38(1), 19–31.
<https://doi.org/10.1007/s11747-008-0131-z>
20. Rane, N., Choudhary, S. P., & Rane, J. (2023). *Hyper-personalization for enhancing customer loyalty and satisfaction in customer relationship management (CRM) systems*. Customer Relationship Management Systems.
<https://doi.org/10.2139/ssrn.4641044>
21. Alfouzan, M., Lu, K., & Ullah, A. (2024). *The effects of artificial intelligence on business marketing quality and factors that positively influence marketing*. International Journal for Quality Research, 18(4), 1021–1034.
<https://doi.org/10.24874/IJQR18.04-07>
22. Bleier, A., & Eisenbeiss, M. (2015). *The importance of trust for personalized online advertising*. Journal of Retailing, 91(3), 390–409.
<https://doi.org/10.1016/j.jretai.2015.04.001>
23. Henker, S., Hernando, C., & Constantinides, E. (2021). *The transformation of data marketing: How an ethical lens on consumer data collection shapes the future of marketing*. Sustainability, 13(21), 11208.
<https://doi.org/10.3390/su132111208>
24. Hart, C. A. (2019). *Evolving perceptions of assortments, image and customer experiences in retail environments*. Loughborough University Research Repository.
<https://hdl.handle.net/2134/23159>
25. KPMG International Cooperative. (2018). *Tomorrow's experience today: Harnessing a customer-first approach*.
<https://assets.kpmg.com/content/dam/kpmg/xx/pdf/2018/06/tomorrows-experience-today-harnessing-a-customer-first-approach.pdf>

26. Tucker, C. E. (2014). *Social networks, personalized advertising, and privacy controls*. Journal of Marketing Research, 51(5), 546–562.
<https://doi.org/10.1509/jmr.10.0355>
27. Ruggiero, T. E. (2000). *Uses and gratifications theory in the 21st century*. Mass Communication & Society, 3(1), 3–37.
https://doi.org/10.1207/S15327825MCS0301_02
28. Blumler, J. G. (2019). *Uses and gratifications research*. In P. V. Rosenthal, V. P. Vos, G. Holtz-Bacha, D. Dimitrakopoulou, M. Gercassi, S. Shagari, & A. Ksolik (Eds.), *The International Encyclopedia of Journalism Studies*. Wiley-Blackwell.
<https://doi.org/10.1002/9781118841576.iejs0032>
29. Kamboj, S. (2019). *Applying uses and gratifications theory to understand customer participation in social media brand communities: Perspective of media technology*. Asia Pacific Journal of Marketing and Logistics, 32(1), 205–231.
<https://doi.org/10.1108/APJML-11-2017-0269>
30. Palmgreen, P. (1984). *Uses and gratifications: A theoretical perspective*. Annals of the International Communication Association, 8(1), 20–55.
<https://doi.org/10.1080/23808985.1984.11678570>
31. Luo, X. (2002). *Uses and gratifications theory and e-consumer behaviors*. Journal of Interactive Advertising, 2(2), 34–41.
<https://doi.org/10.1080/15252019.2002.10722060>
32. Ho, H.-Y., & Syu, L.-Y. (2010). *Uses and gratifications of mobile application users*. In Proceedings of the 2010 International Conference on Electronics and Information Engineering (ICEIE 2010) (Vol. 1, pp. VI-315–VI-319). IEEE.
<https://doi.org/10.1109/ICEIE.2010.5559820>
33. Sebby, J. (2019). *Understanding the effects of personalization behavior on content aggregator use and user satisfaction* (Master's thesis, University of Tennessee). TRACE: Tennessee Research and Creative Exchange.
https://trace.tennessee.edu/utk_gradthes/6101
34. Spang, M. (2017). *An examination of the uses and gratifications of utilitarian and experiential online shoppers* [Master's thesis, Rochester Institute of Technology]. RIT Digital Institutional Repository.
<https://repository.rit.edu/cgi/viewcontent.cgi?article=10533&context=theses>
35. Journalism University. (2023, December 2). *Why we watch what we watch: The uses and gratifications theory*.
<https://journalism.university/media-and-communication-theories/uses-gratifications-theory-media-consumption/>